

Supplementary material.

Digital Object Identifier 10.1109/ACCESS.2026.0429000

Supplement S1 – Protocol and provenance record

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• **ABSTRACT** Supplementary material to "Measuring the Safe Drain Boundary for Durable Backlog Recovery Under a Live Latency Objective". Every item here is cited from the article, which states each item's consequence; this document gives the full record. Text moved from the article is reproduced unchanged.

S1-A. PROVENANCE (FULL TEXT OF THE ARTICLE'S FORMER §IV-G)

Early in the project 137 of 140 run records carried the string "unknown" in place of a commit hash, because the function resolving it swallowed its error on an unborn HEAD. The runner now refuses to start without a resolvable commit. That guard covers the runner's own records — every boundary run carries a commit, a branch, a dirty flag and a start time. It does not cover the standalone saturation tool that measures plateaus, which stamps no provenance of any kind; §V states what that costs and §IX carries it. No run reported in this paper lacks a commit.

One provenance gap remains and is reported rather than closed. The search driver records its results but not its own invocation, so the arguments that produced a cell's probe set are recoverable only by replaying the committed planner against the observed classifications. All seven cells reconstruct exactly, each with no supplied ceiling. One required care: in E1 c10/C1 two SAFE points lie below the floor the descent located, and they carry a run-id prefix identifying them as probes from the replication campaign rather than the original search, which the artefact does record. Replaying the planner on the original campaign's rates alone reproduces that cell's probe set exactly.

What is not recoverable is the command line itself. The gap is therefore in the reproducibility of probe *selection*, not in the measurements: for every cell, all probed rates, all repetitions, all classifications and the terminal 5 rps bracket are retained and used, no run carries an exclusion reason, and none is absent from its cell's analysis — the search driver raises rather than continuing if a run is invalid, so a silently dropped run is not a state it can reach. Adding an argument echo would prevent a recurrence and would recover nothing retrospectively; the driver is left unaltered, because changing a frozen instrument after its campaigns have closed is the practice this section exists to avoid.

S1-B. AMENDMENTS (THE FULL TEXT BEHIND THE ARTICLE'S §IV-H GLOSS)

All ten amendments and record corrections are listed in Table S1 with their commits; the article glosses the seven that later sections cite.

Amendments to the pre-registration, the A7 registered re-analysis, the A8 registered replication and the two record corrections. The two right-hand columns separate registration *before the data existed* from registration *before the analysis was run*; they are different claims and only the first is registration before the data. They do not apply to a record correction, which governs neither, and those rows read N/A rather than being forced into a category they do not belong to.

A7 is a registered re-analysis of an already-collected corpus, not a registration before the data, and the two columns are separated so that the distinction cannot be read the other way.

A8 is neither an amendment nor a re-analysis. It changes no rule and revisits no existing data: it registers a *new* measurement — the window-to-window repeatability of two plateaus the campaign had measured once each — before that measurement was taken, and fixes in advance what each possible outcome would mean, including the outcome that would have removed a claim §V makes about that prediction. It is registered before its own data in the same strict sense as A1 through A3 — subject to the file-timestamp limitation §IX-C records — and appears last only because it postdates the measurement campaign it bears on. Its result is reported in §V-D.

A9 and A10 are a third kind again: record corrections. Each fixes a factual misstatement in the text of an earlier registration, and neither changes a rule, an estimator, a datum or a conclusion. They exist because this registration is immutable once its commit is cited, so a correction takes the form of a dated addendum with the original left standing — and because applying that rule only when a correction is convenient would cost more than the addenda do. A9's error understated what had been registered; A10's makes a reported

TABLE S1. Amendments to the pre-registration, the A7 registered re-analysis, the A8 registered replication and the two record corrections. The two right-hand columns separate registration *before the data existed* from registration *before the analysis was run*; they are different claims and only the first is registration before the data. They do not apply to a record correction, which governs neither, and those rows read N/A rather than being forced into a category they do not belong to.

#	Date	Commit	Change	Predates governed data	Predates governed analysis
A1	2026-09-11	e5bf303	Interval's upper end is the first NON-SAFE point, which may be MARGINAL, not the first UNSAFE point	Yes	Yes
A2	2026-09-11	1fe9de3	A non-SAFE anchor extends the search downward instead of aborting for a hand-chosen anchor	Yes	Yes
A3	2026-09-11	699e105, 25583c7	Live injector moves from a tick-dropping timer to lanes	Yes	Yes
A4	2026-09-11	d5ea89d, ddd428d	Achieved recovery rate measured over the delivery span, not the drain window	Partly — the first boundary's traces (c10/C0) already existed and were recomputed; registered before the remaining E1 cells	Yes
A5	2026-09-12	fe36734	Leading-indicator noise scale changed from mean to median	No	No — chosen with the numbers in view
A6	2026-09-13	c3aee75	A4 is not valid at collapsed points; utilisation is not reported there	No	No
A7	2026-09-14	939902d	<i>Registered re-analysis, not an amendment.</i> The frozen leading-indicator criterion and statistic, re-run over the corrected-harness corpus	No — that corpus already existed	Yes — criterion, statistic and prediction all frozen before the re-analysis
A8	2026-09-14	590d1cc	<i>Registered replication, not an amendment.</i> Repeatability of the two corrected saturation plateaus: ten 60-second windows per arm, median convention, with both arms' readings fixed in advance. No registered constant, no plateau estimator and no analysis path is changed by it	Yes — the twenty windows were collected afterwards	Yes
A9	2026-09-19	9caf476	<i>Record correction, not an amendment.</i> A8's short-arm reading states that neither plateau figure was itself registered as a plateau; 67c448b tabulates both and gates the boundary read on them, so that statement is wrong. No A8 criterion, datum or result depends on it	N/A — governs no data	N/A — governs no analysis
A10	2026-09-19	51026cc	<i>Record correction, not an amendment.</i> A6 reports the E2d collapse factor as $10.3\times$, computed from a midpoint numerator against a SAFE-only denominator; matched on the SAFE side it is $10.5\times$. The denominator, the direction and A6's verdict are unchanged	N/A — governs no data	N/A — governs no analysis

factor 2% larger. The second direction warrants more scrutiny than the first, and §IX gives both in full.

A1 through A3 were registered before any data governed by this registration were collected — an earlier phase of the project had produced data, which is what motivated the protocol — and require no further comment. A8 is described above; the remaining four do.

A4 replaced an estimator, not an instrument. Chronologi-

cally it was registered during E1, after the traces for the first boundary (c10/C0) already existed and before the remaining E1 cells were collected; it was then applied retrospectively to those first traces. The original measure divided an exact backlog by a window padded with the drain detector's confirmation tail, which injected run-to-run noise the rate limiter had not produced. Measuring instead over the span the recovery traffic actually occupied removes that tail. It was

applied to already-recorded traces and required no run to be repeated.

A5 is the one amendment in this document made with the numbers already in view, and it changes an answer. The registration says so in those terms and documents it at length precisely because it is the most attackable step in the leading-indicator analysis. It is reported here on the same basis: a reader should be able to attack it with the evidence in front of them.

A7 attempts to repair that, and does not succeed. A5 froze the statistic on 12 September at 07:31 UTC, having been chosen on the E1 corpus collected five hours earlier — and the leading-indicator analysis reads only that corpus. A7 therefore registered, before running anything, a re-analysis of the same frozen criterion and the same frozen statistic over the corrected-harness corpus, which was collected 34 hours after A5 froze and played no part in choosing it, together with the prediction to be tested. The registration preceded the analysis by commit timestamp.

The prediction did not replicate. The timeout component held cleanly. The ordering component did not: queue depth and live p99 first move at the same probed point in both corrected cells. The registered limitation is what decides the interpretation. E1's safe grids are sampled at 5–10 rps where they are finest; the corrected cells at 60 and 100 rps. A lead of the 10–25 rps that E1 reported is below the corrected corpus's resolution by construction, so that corpus could not have replicated the ordering in either direction. The honest statement is the stronger one: the ordering is unreplicated, and the only independent corpus available was incapable of adjudicating it. §VII reports it as exploratory on those grounds and on A5's chronology independently.

A6 records that A4 is valid at safe points but over-reads at collapsed ones, where the recovery traffic's occupied span is not the drain. Eighteen of twenty unsafe points reported achieved rates above their own saturation plateau — a count, not a magnitude, and therefore not a judgement about resolution. The defect was caught by a cross-check of the estimator against the plateau rather than by inspection, and utilisation is no longer reported at collapsed points. Its consequences for the reported intervals are set out in §IX.

S1-C. REPRODUCIBILITY (FULL TEXT OF THE ARTICLE'S FORMER §IV-I)

The harness, the pre-registration with all six amendments, the A7 registered re-analysis, the A8 registered replication and the A9 and A10 record corrections, every run record and per-request trace — the traces are archived with the dataset rather than held in the repository, so delivery-span quantities are checkable against their artefacts from the repository alone but re-derivable from raw observations only with the deposit — the analysis code, the figure generators and the regression fixture are archived under concept DOI 10.5281/zenodo.22761130. Every reported artefact regenerates byte-identically from the committed data by run-

ning its own script, and a manifest records a SHA-256 for each file together with the commit the package was built from.

That property was verified rather than assumed, and verifying it found a counter-example. The data artefacts reproduced byte-identically throughout; the figures did not, because the plotting library stamps a wall-clock creation time into each file, so two runs over unchanged inputs produced different bytes with identical drawn content. The metadata is now suppressed and determinism is verified over consecutive runs, with the rendered marks confirmed unchanged against the committed versions. The claim as stated held for the artefacts it had been checked against and not for the ones it had not, which is the ordinary way such claims fail.

S1-D. THE CAPACITY MODEL, PICTURED (CUT FROM THE ARTICLE FOR LENGTH)

Fig. S1 pictures the relation equations (??)–(??) state algebraically: the harness staffs its pool on the assumption that a worker turns a request round in exactly the emulated service time, each worker in fact pays a fixed additional cost per request, and because that cost is additive rather than proportional it consumes a larger share of a short service time than a long one. The figure carries no measured result, which is why the article states the relation in equations and keeps the picture here.

S1-E. PER-CELL RESOLUTION IN FULL (THE ARTICLE'S RESOLUTION TABLE, ALL COLUMNS)

The article's resolution table is compacted to the three quantities its argument uses: the boundary bracket, the normalised per-cell resolution and the safe utilisation. Table S2 is the same seven cells with every column the generator produces, so that a reader checking the resolution discipline against the artefacts has the configured and measured capacities, the raw bisection step, the bracket width, the repetition count, the replicate spread and the measured capacity's own range in one place.

S1-F. THE OBJECTIVE AND THE SEARCH (FULL TEXT OF THE ARTICLE'S FORMER §IV-B AND §IV-C)

Reproduced unchanged except that the article's figure keys are written out as figure names, because a supplement cannot number the article's floats.

The objective is evaluated per one-second window over the drain window, on live traffic only. A window is a *latency breach* if live p99 exceeds 250 ms, an *error breach* if the live error rate exceeds 1%, and a *violating second* if either holds. vSLO is the fraction of violating seconds.

Live traffic means injector-direct requests only. Two exclusions apply, both registered in advance, and both are reported here with their exercised extent because an outcome-dependent exclusion in a saturation experiment deserves scrutiny.

Status 429 is an injector-side drop rather than a downstream signal and is excluded from error accounting. The concern this invites is real: an injector that drops more as load rises

What the harness assumes

$c = \lceil C_{\text{config}} \cdot S \rceil$ workers, each turning a request round in exactly S

so capacity $= c/S = C_{\text{config}}$

What a worker actually pays

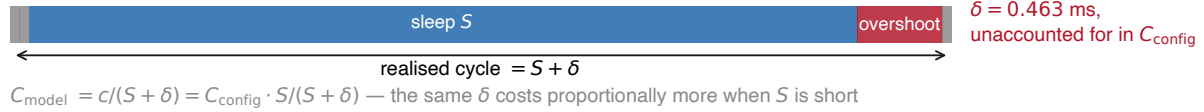


FIGURE S1. Where the unaccounted cost enters. The harness sizes the dependency as $c = \lceil C_{\text{config}} \cdot S \rceil$ workers on the assumption that each turns a request round in exactly S , which makes capacity equal to C_{config} by construction. A worker in fact pays a fixed additional 0.463 ms per request — the pooled saturation-plateau-inferred value — almost all of it the operating system sleeping longer than asked. C_{model} is therefore $c/(S + \delta)$, as in (5), and because the cost is additive rather than proportional it consumes a larger share of a short service time than a long one.

TABLE S2. Per-cell resolution of the boundary estimate, in full: the article's compact table gives the bracket, the normalised resolution and the safe utilisation; this one adds the configured and measured capacities, the raw bisection step, the bracket width, the repetition count, the replicate spread and the measured capacity's own range.

cell	C_{config}	C_{measured}	step (rps)	step / C_{measured}	rate bracket (rps)	width (rps)	n	replicate spread (rps)	$\rho_{\text{eff,safe}}$	C_{measured} range (rps)
E1 c10/C0	2000	1828.6	5	0.0027	[825, 830]	5	15	0.2	0.998	1828.0 – 1829.9 (n = 6)
E1 c10/C1	1400	1280.9	5	0.0039	[275, 280]	5	3	0.0	0.995	1279.1 – 1282.0 (n = 6)
E1 c50/C0	2000	1964.3	5	0.0025	[975, 980]	5	15	1.2	1.000	1962.7 – 1965.3 (n = 6)
E1 c50/C1	1400	1375.7	5	0.0036	[370, 375]	5	3	0.0	0.996	1375.0 – 1378.3 (n = 6)
E2 c10@Q2500	2000	1827.1	5	0.0027	[825, 830]	5	12	0.2	0.999	1825.5 – 1829.5 (n = 15)
E2 c50@Q500	2000	1964.8	5	0.0025	[975, 980]	5	12	1.2	1.000	1963.7 – 1965.7 (n = 15)
E2b C=400	400	392.7	5	0.0127	[190, 195]	5	12	0.0	0.99	392.5 – 393.0 (n = 6)

would starve the downstream and make a run look artificially safe. No run recorded a single injector-side drop — zero across the 171 runs of the seven-cell boundary corpus. The registered $\pm 1\%$ delivery guard checks warm-up rather than the drain, so in-drain delivery was measured separately: the worst sustained deficit among SAFE runs is 0.0099%, a hundredth of one percent, and delivery is marginally *better* in UNSAFE runs than in SAFE ones — the opposite of the direction that would flatter the result.

Seconds flagged by the host-stall detector are excluded from both numerator and denominator, with the unexcluded value recorded as vSLO_{raw} . A second is flagged only when four conditions hold at once: live 1 s p99 at or above five times its rolling healthy baseline, recovery 1 s p99 at or above five times its own, downstream queue depth at or below five requests, and a maximum inter-sample gap in the request stream exceeding 150 ms; each flagged second also contaminates the following five. The third condition is why genuine overload cannot trigger it — at a collapsed point the queue is deep by definition. It never fired: zero seconds were excluded across the 171-run seven-cell boundary corpus, and no classification differs under vSLO_{raw} . It is therefore an untested safeguard,

and what that establishes is not that the detector works but that no result in this paper depends on it.

$\text{vSLO}_{\text{latency}}$, $\text{vSLO}_{\text{error}}$ and $\text{vSLO}_{\text{both}}$ are reported in every table and identify *which* failure mode produced a violation. They do not address sub-threshold degradation, and are not claimed to: a second at p99 = 249 ms is non-violating under all four measures. Continuous live p99 and queue-depth series are therefore retained separately, and it is those that §VII uses. The registration states — before any data — that vSLO alone is not a sufficient safety statistic, because it saturates deep in collapse and is blind to degraded-but-passing traffic at the other edge; the decomposition addresses the first of those, the continuous series the second.

Each probed rate is run $n = 3$ times and classified from those three vSLO values alone:

- SAFE — $\text{vSLO} \leq 0.01$ in all three repetitions.
- UNSAFE — $\text{vSLO} > 0.05$ in at least two repetitions.
- MARGINAL — anything else.

The three classes are exhaustive and mutually exclusive. Repetition disagreement is never averaged, and this applies to rate as well as to vSLO. No single achieved rate is assigned to a probed point: the point retains all three, and the reported

interval spans every achieved ρ observed at each endpoint, so disagreement between repetitions widens the interval rather than being collapsed into a mean or a median. Disagreement across repetitions is itself the signal that an operating point is unstable, which is the property the collapsed regime has, and the mean of three vSLO values is not computed and appears in no table.

The search additionally raises a registered diagnostic whenever the achieved- ρ spread across repetitions at a point exceeds the search resolution, on the grounds that bisection would then be resolving finer than its own instrument. It never fired — zero of forty points across all seven boundary files, endpoints included; the worst point sits at 40% of its own resolution.

Two qualifications, both of which narrow that statement. First, where a single figure is nonetheless required — the per-cell value plotted in the article's collapse figure — it is the maximum achieved ρ across the last SAFE point's repetitions. The choice is semantic before it is statistical: a point is classified SAFE only if all three repetitions satisfy the SAFE criterion, so the maximum is the highest empirically demonstrated safe throughput among repetitions at the terminal SAFE point. It is a defined safe-side quantity rather than the most favourable of several candidates.

It is also immaterial. Recomputing under minimum, maximum, mean and median leaves the monotonicity audit, the identity of the last SAFE point in every cell, and the reported values after resolution-matched rounding all unchanged, because the largest within-point spread anywhere in the campaign is 0.0010, or two requests per second — below both the rounding step and the search resolution.

Second, the scope of that statement has to be given exactly, because two distinct corpora appear in this paper.

The seven cells — four from E1, two from E2, one from E2b; forty probed points, 171 runs — are the boundary result, and they are self-contained. Each cell's denominator comes from that cell's own saturation runs. Every one of the 171 repetitions retains its own achieved rate, so the no-aggregation rule, the spread diagnostic and the aggregator sensitivity all apply to them, and to the article's collapse figure, which plots these cells and only these cells.

The two corrected-harness cells were not produced by this search. They were assembled directly from run records by a separate analysis path that stored one median ρ per point. The per-repetition rates have since been recovered from the 33 retained run records — every record carries the quantities the estimator needs, and the reconstruction reproduces all eleven committed medians exactly — so this is no longer a limitation of the corpus. They support the predict-and-eliminate result of §V, two of the four bars in the article's plateau figure, the signal series of §VII, and the re-analysis registered as A7. They are not members of the seven and contribute to no seven-cell number.

The limitation therefore falls on the corrected cells and not on the boundary result: the repetition-level sensitivity reported above covers §VI's central comparison, and does not

cover §V's brackets, the article's plateau figure's corrected bars, §VII or A7. The recovered per-repetition rates change nothing: the predict-and-eliminate verdict is identical under all four aggregators, and the A7 ordering is unchanged. Their per-point spread of 2.1 to 7.1 rps would flag against the registered 5 rps constant, but those searches resolved to 55 and 65 rps rather than 5, so against the resolution actually achieved the spread is eight to ten times smaller than the step — more headroom than anywhere in the seven-cell corpus. The one difference that remains is estimator: the seven measure over the delivery span under A4, the corrected cells over the drain window as measured. The article's plateau figure places both side by side, which is legitimate because predicted against measured plateau is a direct throughput comparison with no ρ estimator involved, but its caption must say so.

The initial bracket is constructed rather than assumed. A ceiling may be supplied; if it is not, the first candidate is the anchor raised by ten percent, rounded to 5 rps, and stepped again while it continues to classify SAFE. Every candidate — supplied or stepped — is freshly probed at the campaign's own repetition count before it is used, on every harness and without exception, so every rate appearing in a boundary file was actually run. A supplied ceiling that classifies SAFE is demoted to the floor and the upward search restarts from it. The downward search has the 5 rps resolution floor as its terminating guard; the upward search has no probe budget and no equivalent guard, an asymmetry that did not arise in this campaign but is a defect in the procedure rather than a property of the results.

The search then bisects between the last-SAFE anchor and the first-non-SAFE ceiling. Midpoints are rounded to the nearest 5 rps. A SAFE midpoint becomes the new floor; an UNSAFE or MARGINAL midpoint becomes the new ceiling, so marginal behaviour lies inside the reported interval rather than below it. The anchor is probed first and, if it does not classify SAFE, the search steps downward rather than assuming a floor (amendment A2). Anchors inherited from an earlier phase do not count, because those points were measured on the defective harness.

The search stops when the bracket is 5 rps wide, and the registration states that 5 rps is the resolution floor and that no claim is made below it.

Bisection presupposes that safety is non-increasing in offered load, and the search terminates at the first non-SAFE classification. The two misclassification errors are not symmetric in their consequences. A spuriously non-SAFE point truncates the search conservatively, moving the reported last-SAFE point downward and therefore *away* from capacity — it costs the headline rather than supporting it. The consequential error is the opposite one: a spuriously SAFE point above the true boundary moves the safe-side estimate upward, toward the claim the paper makes. Requiring all three repetitions to satisfy the SAFE criterion, against two of three for UNSAFE, makes that classification deliberately the stringent one. The monotonicity assumption is nonetheless audited rather than assumed, and the audit needs its own rule, because a point

retains three achieved rates and cannot be ordered by all three at once. The audit therefore proceeds at the point level using a single achieved rate per point, and was repeated under each of the four candidate aggregators — minimum, maximum, mean and median. The rank order of points within a cell is identical under all four, so the audit reads the same input in every case. No inversion occurred in any of the seven cells, under any aggregator, where an inversion means a higher-rate point classifying safer than a lower-rate one.

Two facts bound what that audit establishes. First, it can speak only for the forty points that were probed; bisection does not sample what it does not choose. Second, no point in the study ever classified MARGINAL — the forty points divide as twenty SAFE and twenty UNSAFE. The MARGINAL class, and amendment A1 which governs its treatment as a ceiling, were therefore never exercised on any reported result. They are retained in the protocol as registered, not as machinery that shaped an outcome.

One bracket is weaker than the rest and is named rather than averaged into the others. In E1 c10/C0 the last SAFE point reports three zero vSLO values and the first UNSAFE point reports 0.077, 0.000 and 0.135 — one replicate at exactly zero. Under the registered rule this is UNSAFE, since two of three exceed 0.05. The disagreement is not a defect in the classification but the instability the registration anticipated at an unstable operating point, and it is reported as such.

S1-G. ADMISSION AND THE CONFIGURED-CAPACITY ENDPOINT (FROM THE ARTICLE'S FORMER §III-C)

The article keeps equation (??) and the cancellation argument; the implementation detail and the endpoint history are here, unchanged.

Under the graceful profile the admission limit is $\text{queueCap} = 50 \cdot c$; and under the cliff profile it is $2 \cdot c$, returning an immediate rejection once full. The queue channel itself is allocated at four times the admission limit; admission is enforced by a token pool rather than by channel capacity, so the limit can be changed without reallocating.

The time to traverse a full queue is the admission limit divided by the service rate. Under the intended occupancy model the service rate is c/S and the traversal time is $50 \cdot S$ — 250 ms at $S = 5$ ms. Under the corrected occupancy model of §III-D the service rate is $c/(S + \delta)$ and the traversal time is $50 \cdot (S + \delta)$. In both cases the worker count cancels. The admission rule therefore cannot by itself produce a capacity-dependent or concurrency-dependent queue-delay scale, under either the intended model or the corrected one — a point §VIII returns to, since one of the candidate explanations proposed exactly that mechanism.

One separation is deliberate and load-bearing. The configured capacity parameter is exposed only on an administrative endpoint, which the consumer never reads. The harness was built so that a recovery controller could not rely on the configured capacity parameter and would have to operate from observations instead. The results later justify distrust of that configured value, while not determining whether a validated

estimate should be supplied offline or inferred online — a distinction §I leaves open and this paper does not settle. The decision was taken before any measurement, for a reason narrower than the one that ultimately justified it.

It is worth recording that the endpoint's response field is named `trueCapacity`,

and the specification describes it as exposing true capacity. It does not: it returns C_{config} . The name is itself a residue of the assumption this paper falsifies, and it is preserved unaltered in the archived artefact.

S1-H. THE CORRECTED CORPUS IN DETAIL (THE FULL TEXT BEHIND THE ARTICLE'S §VII GLOSS)

Reproduced unchanged except that the article's figure key is written out as a figure name.

S1-H.1 THE CORRECTED CORPUS: WARNING WITHOUT ORDERING

The corrected corpus answers differently in each of its two cells, and the two answers together are the result.

In the corrected long-service cell there is no warning room at all. Mean queue depth, all three live latency percentiles and mean in-flight requests first cross at r_l 1210, which is the last safe point. On that cell the leading-indicator framing has nothing to lead with.

In the corrected short-service cell there is warning room but no resolved ordering. Mean queue depth, live p99, live p90 and mean in-flight all first cross at r_l 1075, with the last safe point at 1185 still ahead of them — one probe point, roughly 110 requests per second of safe operating range after the first crossing. Only live p50 crosses with no room, and only timeout rate never crosses. But those four cross *together*: what is absent in this cell is the ordering, not the warning.

The corrected corpus therefore does not reproduce the queue-before-latency ordering at its available resolution. It does not show that no metric gives advance warning. Those are different claims and this section keeps them apart.

One further observation belongs here, with its quantity named. In the short-service cell the *mean* drain queue depth never reaches its DEEP threshold anywhere inside the safe range, peaking at 23.1 requests against 50. That is a statement about the mean; the article's signal figure plots the queue *peak*, which behaves differently and is discussed below.

A possible explanation suggests itself, and is offered as a hypothesis rather than as a finding. The corrected corpus samples coarsely near the boundary — 60 and 100 requests per second between safe points, against E1 margins of 10 to 25 — so it may leave too little resolution to distinguish the relative *onset* of queue depth and live latency even where both give advance warning. Under that hypothesis a lead of E1's size would disappear below this corpus's sampling resolution. Whether the original separation was itself related to the calibration error remains untested.

This is not a fourth retracted finding, and the evidence here does not support making it one. It is a hypothesis with a named experiment: a 5 requests per second sweep across the top 50

of each corrected cell's safe range, about sixty runs at $n = 3$, which would resolve a lead of E1's size in either direction. §IX records it as the study that would settle the question.

S1-H.2 WHAT THE FIGURE SHOWS, AND WHY IT IS UNAFFECTED

the article's signal figure plots drain queue peak and live tail latency across the corrected short-service cell as utilisation approaches its boundary. Both rise across the sampled safe range — the queue peak from 7 requests to 111, live p99 from 8 ms to 58 — and both then change sharply at the transition between the final SAFE point and the following non-SAFE one: the queue reaches its cap of 500 and live p99 rises to 508 ms. That interval is 55 requests per second, not 5. The corrected-harness searches did not resolve at the seven-cell corpus's 5 rps step; §IV records their achieved resolutions as 55 and 65 rps, and they are not members of the seven-cell boundary corpus.

The figure is unaffected by the downgrade in §VII-B, and the reason is worth stating. Its pipeline uses no noise statistic anywhere — it reads utilisation, queue peak, live p99 and the classification directly — so it never depended on the quantity A5 chose or A7 re-examined. A figure that survives a downgrade because it never rested on the downgraded thing is evidence about the figure, not a coincidence.

What §VII does not claim is that latency warns last, or that either signal is the right one to build a controller on. §II establishes live latency as a literature-grounded comparator, which is what makes these observations about a signal the literature already uses. This section reports only that on this harness, at this resolution, timeout rate never warns; the queue-depth lead observed on E1 is unreplicated and unresolvable on the available data; and on the corrected corpus some observables do give advance warning in one cell, but queue depth does not lead live p99 at the resolution available.

S1-I. TWO DOCUMENTATION FAILURES, AND TWO HARNESS DEFECTS

Two documentation failures belong here, recorded for what caught them; the full text is the article's, cut for length. An incorrect reading of the corrected-corpus figure propagated through report prose, a caption, the title drawn inside the figure itself and an early section draft, because each layer was written from the previous summary rather than from the data, while the registered analysis's own table contradicted it throughout. Separately, a summary generalised one evidentiary pattern across three candidate explanations; generating the table meant to display them, with every quotation asserted against its source, showed that one had been refuted before calibration and another had no like-for-like post-calibration test. Neither is an experimental finding, and the lesson is the control rather than the error: for a claim about several items at once the authority is a generated table whose cells are checked against their sources, not the prose in between.

Two harness defects belong with them, moved here from the article's §IX-E and reported as evidence that the check-

ing regime operated rather than as incidental. A spin-wait admission profile was found to amplify a transient rejection into a sustained collapse, and was replaced. The overhead record's own descriptive field contradicted the arithmetic of the record it was written into, in every record it wrote, until it was corrected at the source and in all 43 affected records — after which every consuming artefact regenerated byte-identically.

...